

GradeSense Deployment Guide

Local, Docker, Render, environment, WebSocket, and operational verification

Curated, implementation-linked documentation. Code examples are selectable text, never screenshots.

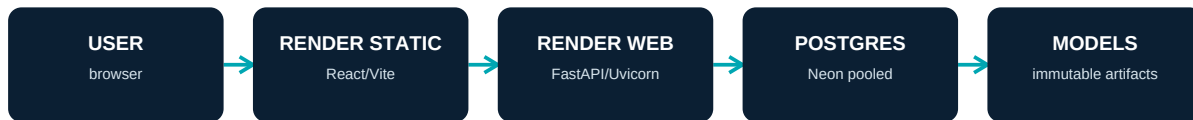
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1. Deployment Topology



The frontend and backend deploy independently. The frontend receives `VITE_API_URL` at build time. The backend receives `DATABASE_URL` and allowed frontend origin at runtime.

2. Backend Setup

```
cd backend
python -m venv .venv
.venv\Scripts\activate
pip install -e ".[dev]"
alembic upgrade head
uvicorn app.main:app --reload
```

Application lifecycle

```
configure_logging(settings.log_level)
logger = get_logger(__name__)

@asynccontextmanager
async def lifespan(_: FastAPI) -> AsyncIterator[None]:
    with SessionFactory() as session:
        IntelligenceService(settings, session).ensure_ready()
        ModelRegistryService().bootstrap_existing(session, settings)
        AuditService().record(session, "application_startup", "application")
    stream = get_streaming_service(settings)
    await stream.start()
    logger.info("application_started", extra={"environment": settings.environment})
    yield
    await stream.stop()
    with SessionFactory() as session:
        AuditService().record(session, "application_shutdown", "application")
    dispose_engine()
    logger.info("application_stopped")

def create_application() -> FastAPI:
    application = FastAPI(
        title=settings.app_name,
        version=settings.app_version,
        description="AI decision-support API for industrial paper grade transitions.",
        docs_url="/docs",
        redoc_url="/redoc",
        openapi_url="/openapi.json",
        lifespan=lifespan,
```

3. Frontend Setup

```
cd frontend
npm ci
set VITE_API_URL=http://localhost:8000
npm run dev
```

The production static build requires VITE_API_URL. Missing configuration throws during client initialization instead of falling back to an unsafe implicit origin.

```
RelationshipDiscovery,
DemoSeedResult,
} from './types'

const API_URL = (import.meta.env.VITE_API_URL ?? '').trim().replace(/\/$/, '')
if (!API_URL) {
  throw new Error('VITE_API_URL must be configured before building or starting GradeSense.')
}

export class ApiError extends Error {
  constructor(
    message: string,
    public status?: number,
    public details?: unknown,
  ) {
    super(message)
  }
}

async function request<T>({
  path: string,
  options?: RequestInit,
  timeoutMs = 30_000,
}): Promise<T> {
  const controller = new AbortController()
```

4. Environment Variables

Variable	Service	Purpose
DATABASE_URL	Backend	Pooled PostgreSQL URL
GRADESENSE_DATABASE_URL	Backend	Application SQLAlchemy override
GRADESENSE_CORS_ORIGINS	Backend	Allowed frontend origins
GRADESENSE_ENVIRONMENT	Backend	Runtime environment
GRADESENSE_LOG_LEVEL	Backend	Structured logging level
VITE_API_URL	Frontend	Backend origin compiled into client

Secrets belong in the deployment platform. Do not commit database credentials or production tokens.

5. Docker Deployment

```
docker compose config --quiet
docker compose up --build
docker compose ps
docker compose logs -f backend
```

Container	Responsibility
frontend	Nginx serves the Vite production build
backend	Uvicorn serves FastAPI and WebSockets
database	PostgreSQL for concurrent local integration
models/data	Mounted or copied as read-only runtime inputs

6. Render Blueprint

```

services:
  - type: web
    name: gradesense-api
    runtime: python
    plan: free
    buildCommand: >
      pip install --upgrade pip &&
      pip install ./backend &&
      cd backend &&
      export GRADESENSE_DATABASE_URL="$DATABASE_URL" &&
      alembic upgrade head
    startCommand: >
      cd backend &&
      export GRADESENSE_DATABASE_URL="$DATABASE_URL" &&
      export GRADESENSE_CORS_ORIGINS=["https://gradesense-4weh.onrender.com"] &&
      exec uvicorn app.main:app --host 0.0.0.0 --port "$PORT"
    healthCheckPath: /health
    maxShutdownDelaySeconds: 30
    autoDeployTrigger: commit
    envVars:
      - key: PYTHON_VERSION
        value: 3.13.5
      - key: GRADESENSE_ENVIRONMENT
        value: production
      - key: GRADESENSE_LOG_LEVEL
        value: INFO
      - key: DATABASE_URL
        sync: false

  - type: web
    name: gradesense
    runtime: static
    buildCommand: cd frontend && npm ci && npm run build
    staticPublishPath: ./frontend/dist
    autoDeployTrigger: commit

    envVars:
      - key: VITE_API_URL
        value: https://gradesense-api.onrender.com

    routes:
      - type: rewrite
        source: /*
        destination: /index.html

```

The backend build installs the package, exports the database URL for Alembic, applies migrations, and starts Uvicorn on Render's assigned PORT. The static service publishes frontend/dist.

7. WebSocket and Operational Verification

```
# REST readiness
curl https://<api-host>/health

# Browser WebSocket target
wss://<api-host>/ws/dashboard
```

Check	Expected result
Health	ready with artifact and database status
Prediction	POST /predict returns persisted UUID
Forecast	12-step trajectory and crossing risk
WebSocket	sensor and inference events without console errors
Database	Alembic head applied
Audit	startup and administrative actions recorded

Production constraints

- Use PostgreSQL, not SQLite, for concurrent production traffic.
- Place the service behind enterprise identity and authorization.
- Use a shared broker before adding multiple API replicas.
- Keep model artifacts immutable and checksum-validated.